

Semiconductors:

T1, T2	- BC557 pnp transistor
T3-T5	- BC547 npn transistor
D1	- 1N4001 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2	- 150-kilo-ohm
R3, R4	- 10-kilo-ohm
R5, R6, R10	- 1.5-kilo-ohm
R7	- 15-kilo-ohm
R8	- 1-mega-ohm
R9	- 100-kilo-ohm
R11	- 4.7-kilo-ohm

Capacitors:

C1 - 10 μ F, 16V electrolytic
C2 - 0.01 μ F ceramic disk

Miscellaneous:

LDR1, LDR2	- Light-dependent resistors
CON1	- 2-pin terminal connector
S1	- On/off switch
RL1	- 6V, 1C/O relay
PZ1	- Piezo buzzer
	- 6V battery
	- Two laser torches
	- 2-pin connector
	- Basic mobile phone
	- Eight small mirrors

Construction and testing

The circuit works off a 6V battery or power supply. Refer Fig. 4 for the



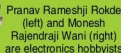
proposed installation of laser beams and mirrors for the security system.

Switch S1 to off position during installation of the laser beams and LDRs.

The project requires two mobile phones—one old basic feature phone, which will be connected to the circuit, and the other could be the mobile phone you may already be using.

Normally, button 1 on a feature phone is assigned for speed dial function. Open the cover of the dial pad of this feature phone. Solder two wires on the PCB at 1 (Fig. 5). Extend these wires to CON2 in the circuit.

You can install the two laser beams at a slight distance from each other, depending on what you want to guard against. For example, to guard your farm against cattle, keep the two laser beams about 90cm apart, so that the animal interrupts both the laser beams when it crosses the gate of your farm. As soon as animal enters your farm, this system will alert you, and you can take instant action. With the two laser beams kept 90cm apart, a human being crossing the gate of your farm will not produce an alarm or send an alert to your mobile phone, thus preventing a false alarm. **EFY**



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